A NATIVE-REFERENCE PHONE-CLASS GEOMETRY FOR SECOND-LANGUAGE PRONUNCIATION ANALYSIS

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ABSTRACT

Automatic speaking assessment systems can provide holistic proficiency scores, but often lack interpretable measures that characterize pronunciation quality. We propose a native-reference phone-class geometry for measuring second language (L2) pronunciation deviation without requiring pronunciation labels, read-aloud prompts, or matched recordings of the same text from native and L2 speakers. Given a native speech corpus, we average frame-level self-supervised representations for each context-dependent phone-class and use singular value decomposition (SVD) to derive a compact native-reference coordinate system. For each L2 utterance, we compute the corresponding averages and project them into the native-reference space. We then demonstrate that the distances between L2 and native-reference coordinates for matched phone-classes show consistent negative correlations with holistic speaking proficiency on the Dev subset of the Speak and Improve Corpus 2025 (Spearman's $\rho = -0.53$) and with pronunciation quality on the learner subset of the English Read by Japanese Students dataset ($\rho = -0.34$). These findings suggest that the proposed geometry captures acoustic-phonetic information relevant for proficiency rating while remaining applicable to spontaneous L2 speech without matched native recordings.

***Index Terms*—** second-language speech, pronunciation assessment, automatic speaking assessment, self-supervised speech models

1. INTRODUCTION

Automatic speaking assessment (ASA) aims to evaluate second language (L2) speech and has growing applications in language proficiency rating and learning. Existing systems combine information from several dimensions, including fluency, pronunciation, lexical use, and grammar, to estimate holistic speaking proficiency [1, 2]. Among these dimensions, pronunciation is usually the most challenging to address due to fine-grained acoustic variations over time. While holistic scores summarize overall speaking ability, understanding the relationship between individual speech properties and these scores is important for interpreting assessment outcomes and developing targeted feedback for language learners. Acoustic representations corresponding to phonetic segments provide one way of characterizing variation in L2 speech. Previous work has shown that distances derived from phone-specific acoustic representations can contain information associated with overall speaking proficiency, including for spontaneous speech [3]. From a speech processing perspective, a key source of information derives from the acoustic realization of words, and more specifically subword units, by L2 speakers. This motivates the development of reliable pronunciation assessment models that can capture the relevant acoustic cues while remaining robust to L2 speech variability. Methods such as Goodness-of-Pronunciation (GoP) are well suited to constrained read-aloud settings, where the expected prompt is known and phone-level pronunciation evidence can be derived from acoustic model likelihoods or posterior probabilities. Applying such approaches to spontaneous L2 speech is more challenging, as disfluencies and pronunciation variability can make automatic transcription and phone alignment less reliable, while acoustic models may also be affected by mismatch between their training data and L2 speech. This issue is particularly relevant for frame-wise models with blank-based label topologies, such as connectionist temporal classification (CTC), where peaky output distributions [4] and the role of the blank label complicate the interpretation of frame-level posterior evidence. This motivates a complementary question: rather than deriving pronunciation evidence from likelihood or posterior distributions over phone labels, can we instead analyze the continuous phone representations directly?

Recent research has examined the phonetic information encoded by self-supervised speech models [5, 6, 7, 8], showing that representations extracted from phone-aligned segments capture context-dependent phone information [9, 10] and extend beyond the corresponding phone to reflect broader contextual information [11]. Nevertheless, these findings do not directly provide an interpretable metric for L2 pronunciation assessment. Distance-based approaches have also been explored using dynamic time warping alignments between L2 and native speech [12, 13, 14]. Such methods, however, require matched recordings of the same text from native and L2 speakers and can be sensitive to disfluencies, which are common in L2 speech.

We propose a native-reference phone-class geometry that is applicable to spontaneous L2 speech and does not require matched linguistic content between native and learner utterances. Instead of comparing whole utterances, we first construct context-dependent phone-class *centroids* by averaging frame-level self-supervised representations over their corresponding segments. By performing a low-rank singular value decomposition (SVD) of the native centroid matrix, we define a low-dimensional native-reference basis that captures dominant variation among native phone-class realizations. For each L2 utterance, we project its phone-class centroids into the native projection basis, yielding L2 phone-class coordinates. We then measure the pronunciation deviation as the distance between L2 and native coordinates for matching phone-classes, using different distance measures. In particular, we do not train a model using the centroids or the distances to predict the pronunciation or proficiency level. Instead, we evaluate whether the raw native-reference distances are monotonically associated with the available human ratings: holistic speaking proficiency for Speak and Improve Corpus 2025 (S&I) and pronunciation quality for English Read by Japanese Students dataset (UME-ERJ).

This paper extends our earlier work [15] and further assesses the behavior and validity of our proposed method by evaluating spontaneous speech from S&I and read-aloud speech from UME-ERJ across several experimental settings. Since comparable results are not available in prior work, we designed a series of baseline experiments. We then test whether the observed association with proficiency (1)

is robust to the choice of native-reference corpora across TIMIT, 4h subset of LibriSpeech 960h (LBS-4h), and native subset of UME-ERJ, (2) transfers to the read-aloud UME-ERJ pronunciation setting, and (3) depends on methodological choices such as projection space or alignment quality. Our native-reference distances show consistent negative correlations with proficiency, suggesting that the proposed geometry captures relevant acoustic information without requiring read-aloud speech or matched recordings of the same text from native and L2 speakers.

2. PROPOSED METHOD

Let $H^\ell \in \mathbb{R}^{T\times D}$ denote the sequence of encoder representations extracted from layer ℓ of a self-supervised speech model. Given the phonetic transcription of a spoken word sequence, we define context-dependent *phone-classes* by augmenting each center phone with left context for diphones and both left and right context for triphones. We denote by $\mathcal{P}$ the set of resulting phone-classes. Given an aligned phone-class sequence $\{a_t\}_{t=1}^{T}$ obtained by forced alignment, we define $\mu(p) = \frac{1}{|\mathcal{T}_p|}\sum_{t\in\mathcal{T}_p} h_t$ as the phone-class *centroid* for class $p \in \mathcal{P}$, where $\mathcal{T}_p \coloneqq \{t : a_t = p\}$. The centroids are obtained by using feature-slicing [5] with two averaging strategies. In *center-unit* averaging, only the frames aligned to the center phone are used. In *full-unit* averaging, all frames belonging to the complete context-dependent unit, including its contextual phones, are pooled. Each centroid summarizes the typical realization of a phone in a specified phonetic context encoded by the self-supervised representation. The resulting phone-class centroids are stacked as the rows of a *centroid matrix* $M \in \mathbb{R}^{|\mathcal{P}|\times D}$.

2.1. Native-Reference and Phone-Class Coordinates

The native centroid matrix M_{nat} is constructed by considering the set of phone-classes derived from the transcriptions of a native speech corpus. Since the rows of M_{nat} are pooled across all native utterances, each centroid for a phone-class $p \in \mathcal{P}_{\text{nat}}$ represents a corpus-level estimate rather than a speaker or sentence-specific realization. We introduce a global native mean vector $\bar{\mu}_{\text{nat}} = \frac{1}{|\mathcal{P}_{\text{nat}}|}\sum_{p\in\mathcal{P}_{\text{nat}}}\mu(p)$, and denote

$$\bar{M}_{\text{nat}} = M_{\text{nat}} - \mathbf{1}\bar{\mu}_{\text{nat}}^\top$$

as the centered centroid matrix. In order to obtain a native-reference phone-class coordinate system, we perform rank-K truncated SVD of the native centered centroid matrix

$$\bar{M}_{\text{nat}} \approx U\Sigma V^\top, \quad U \in \mathbb{R}^{|\mathcal{P}_{\text{nat}}|\times K},\ \Sigma \in \mathbb{R}^{K\times K},\ V \in \mathbb{R}^{D\times K},$$

where $U\Sigma V^\top$ is the best possible rank-K linear approximation of the centered native phone-class mean matrix under Frobenius norm [16]. The *projection basis matrix* V contains orthonormal directions spanning a K-dimensional native-reference subspace of the original self-supervised feature space, capturing dominant variation among centered native phone-class mean vectors. The first direction v_1 captures the largest variation among native phone-class centroids, while subsequent directions capture the largest remaining orthogonal variations. The rank K controls how much of the native acoustic-phonetic variation is retained in the coordinate system.

We now define $C_{\text{nat}} = U\Sigma \in \mathbb{R}^{|\mathcal{P}_{\text{nat}}|\times K}$ as the *native-reference coordinate matrix*. Its rows contain the coordinates of the phone-class mean vectors in the basis defined by $\{v_k\}_{k=1}^{K}$, with each row corresponding to one phone-class and each column to one SVD direction. While the columns of V define the native-reference basis, the matrix C_{nat} contains the coordinates of the native phone-class centroids in this basis. For an L2 utterance r and its self-supervised feature sequence $H_r^\ell \in \mathbb{R}^{T_r\times D}$, we build the L2 centroid matrix $M_{\text{L2}}^r \in \mathbb{R}^{|\mathcal{P}_{\text{L2}}^r|\times D}$ in the same way as for M_{nat} with the difference that now $\mathcal{P}_{\text{L2}}^r$ is the set phone-classes appearing in the transcript of the L2 utterance r. Unlike the native centroid matrix, M_{L2}^r is utterance-level, i.e., it contains only the phone-classes observed in the current L2 utterance. The corresponding *L2 coordinate matrix* is then calculated by projecting M_{L2}^r into the native SVD subspace V, after subtracting the native global mean $\bar{\mu}_{\text{nat}}$, resulting in $C_{\text{L2}}^r = (M_{\text{L2}}^r - \mathbf{1}\bar{\mu}_{\text{nat}}^\top)V$.

2.2. Native-Reference Distance and Evaluation

For each utterance r, we restrict the comparison to the phone-classes shared by the native and L2 representations, $\mathcal{P}_r^\star = \mathcal{P}_{\text{nat}} \cap \mathcal{P}_{\text{L2}}^r$, yielding $C_{\text{nat}}^\star$ and $C_{\text{L2}}^{r,\star}$. For each shared phone-class p, we measure the deviation by $\delta_r(p) = d_m(C_{\text{nat},p}^\star, C_{\text{L2,p}}^{r,\star})$, where d_m is Euclidean, cosine, or diagonal Mahalanobis distance. The utterance-level distance aggregates $\delta_r(p)$ over $p \in \mathcal{P}_r^\star$. This ensures that each comparison is made between the same phone-class in native and L2 speech, while using a fixed native mean and basis for all L2 utterances. We evaluate the association between the resulting utterance-level distance and proficiency or pronunciation scores using Spearman's ρ [17]. We also report Pearson's r [18] in our final evaluation as a complementary linear association measure. Since lower distances indicate greater similarity to the native-reference coordinates and higher scores indicate higher proficiency, negative correlations indicate that native-reference distance decreases with proficiency.

3. EXPERIMENTS

3.1. Setting

To construct our native-reference coordinates, we selected 4 hours of speech from each of TIMIT [19], LibriSpeech 960h [20], and UME-ERJ [21]. TIMIT comprises 630 speakers across its three subsets. For comparability, we sampled 630 speakers from LBS-4h, while UME-ERJ contains only 20 speakers. While all three native-reference corpora contain read speech, they differ in style: LBS-4h is derived from audiobooks, UME-ERJ consists of single sentences read under controlled conditions, and TIMIT provides phonetically balanced speech with ground truth phone segmentation. For L2 speech, we use Train and Dev partitions of the S&I [22, 23] together with the learner subset of UME-ERJ. For the S&I Train, only utterances with manual transcriptions are used. For both learner and native subsets of UME-ERJ, we only considered the *sentences* partition. The correlation coefficients are computed between our proposed distances and the proficiency scores for S&I, and between our proposed distances and the pronunciation scores for UME-ERJ. For S&I, each subset is divided into four proficiency-evaluation tasks. We compute the correlation coefficient for each task and report the average, following common practice. For UME-ERJ, each sentence is read by several native and learner speakers, and we follow prior work [14] for score aggregation.

We selected the best distance measure, encoder layer, and SVD rank using S&I Train and UME-ERJ learner, with LBS-4h serving as the native-reference corpus. We use speech representations from layers 6 and 12 of WavLM Base+ encoder [24] for S&I and UME-ERJ, respectively. As the test scenario, we report results on S&I dev-LBS-4h and learner-native UME-ERJ using the best configuration, consisting of SVD rank 32, phone-class (S&I: triphone, UME-ERJ: diphone), and averaging strategy (S&I: full-unit, UME-ERJ: center-unit). We use the ARPAbet phones and merge stressed phones into a unique class for a total of 39 phones and one silence label. For all corpora except TIMIT, which provides ground truth alignments, phone segmentations are obtained by forced alignment using a Conformer-based factored Hidden Markov Model (HMM) [25] trained on the small Loquacious subset [26]. For our baselines using Voice Activity Detection (VAD), speech regions are obtained by frame-level major-

Table 1: Non-acoustic baselines. We report Spearman correlation ρ for S&I Train, S&I Dev, and UME-ERJ learner using LBS-4h native-reference corpus. Higher positive correlations indicate better agreement with proficiency level. $\mathcal{P}^\star$ denotes the matched context-dependent phone-classes between native and L2 coordinates in our proposed method.

Source	Feature	S&I		UME-ERJ
		Train	Dev	
Raw	# Words	0.55	0.57	−0.20
	Speaking rate	0.54	0.52	0.13
$\mathcal{P}^\star$	Diphones	0.60	0.62	−0.23
	Triphones	0.60	0.62	−0.22

Table 2: Acoustic baselines for similar evaluations and datasets as in Table 1. Here, lower negative correlations indicate better agreement with proficiency. The silence ratio is calculated by considering both voice activity detection (VAD) and silence segments from the alignment. Encoder averaging and dynamic time warping (DTW) apply only to UME-ERJ, which provides parallel learner–native recordings.

Source	Feature	S&I		UME-ERJ
		Train	Dev	
Silence	VAD	−0.36	−0.41	0.10
	Alignment	−0.60	−0.57	−0.07
ASR	WER	−0.22	−0.20	−0.24
Raw Encoder	Mean	N/A		−0.20
	DTW			−0.50

ity voting over three systems (Pyannote [27], Silero VAD [28], and FireRedVAD [29]), after which pauses shorter than 25 milliseconds are merged into the surrounding speech. We used Lhotse [30] for data preparation and RASR [31] for forced alignment. We provide the source code of our experiments [32].

3.2. Baselines

We first report baseline controls that may correlate with speaker proficiency but do not rely on acoustic information or distance measures. Table 1 shows that, on S&I, word count, speaking rate, and matched phone-class coverage $|\mathcal{P}_r^\star|$ correlate strongly with holistic scores ($\rho \approx 0.5$-0.6). This is consistent with spontaneous speech where higher-proficiency speakers produce more fluent and longer responses, while using a broader vocabulary range, as reflected in greater matched phone-class coverage. In contrast, the same features show weak or negative correlation on UME-ERJ, which consists of read-aloud speech and is rated specifically for pronunciation quality. Thus, these features provide important information for interpreting whether native-reference distances capture acoustic variation beyond phonetic coverage. In Table 2, we also report acoustic baselines capturing fluency, automatic speech recognition (ASR) accuracy, and DTW-based acoustic similarity between speakers producing matched sentences. On S&I, silence shows a strong negative correlation, especially when derived from an alignment ($\rho \approx -0.60$), confirming that the proficiency level is strongly affected by pauses. In contrast, these features are weak on UME-ERJ, where ratings target pronunciation in read speech. WER shows only moderate negative correlations on both datasets. The raw-encoder DTW baseline is strong on UME-ERJ ($\rho = -0.50$), but it requires matched recordings of the same text from native and L2 speakers and is therefore not applicable to S&I. These baselines provide reference points for interpreting our native-reference distances, which do not require matched recordings.

3.3. Effect of Phone Alignment

Our proposed distances do not rely on likelihoods or posterior probabilities from an acoustic model, nor on GoP scores. However, the computation of the phone-class centroids still depends on phone-level segmentations. Therefore, we examine the robustness to random phone-boundary perturbations and verify whether the distance-proficiency correlation is preserved across different phone-class types and averaging strategies. When computing the L2 phone-class centroids, we randomly perturb the phone boundaries by shifting them with zero-mean Gaussian noise of varying standard deviation, while keeping the phone sequence fixed. We evaluate diphone and triphone phone-classes with center-unit and full-unit averaging on S&I Train, and calculate the distances with respect to phone-class coordinates obtained from LBS-4h native-reference corpus. As shown in Table 3, increasing the magnitude of the boundary perturbation generally weakens the negative association with proficiency, although its effect depends on the phone-class type and averaging strategy. Triphone representations are more robust than diphones, and full-unit averaging retains stronger negative correlations than center-unit averaging as the perturbation increases. In particular, the two triphone strategies are nearly identical without perturbation, whereas full-unit averaging remains more robust under larger boundary perturbations.

Table 3: Effect of random boundary shift in milliseconds (ms) for different phone-classes and averaging strategies (Avg.) using the LBS-4h native corpus. Values are Spearman ρ of the cosine distance on the S&I Train, averaged over three random seeds.

Class	Avg.	Random Boundary Shift [ms]					
		0	20	40	60	80	120
Diphone	Center	−0.46	−0.45	−0.42	−0.36	−0.30	−0.17
	Full	−0.51	−0.50	−0.47	−0.41	−0.33	−0.19
Triphone	Center	−0.55	−0.55	−0.52	−0.48	−0.42	−0.29
	Full	**−0.56**	**−0.56**	**−0.54**	**−0.51**	**−0.46**	**−0.34**

Table 4: Spearman ρ for three native-reference (Native-Ref) corpora evaluated on S&I Train and UME-ERJ learner subset using three distance metrics: Euclidean (Euc), Mahalanobis (Maha), and cosine distance. UME-ERJ native is held out for testing, see Table 6.

Native-Ref Corpus	S&I Train			UME-ERJ		
	Euc	Maha	Cosine	Euc	Maha	Cosine
TIMIT	−0.44	−0.43	−0.51	−0.25	−0.26	−0.13
LBS-4h	−0.52	−0.51	**−0.56**	−0.27	**−0.28**	−0.13
UME-ERJ	−0.35	−0.32	−0.48	–		

3.4. Different Coordinate Constructions

Table 4 shows that the proposed distances are consistently negatively correlated with proficiency across all native-reference and L2 evaluation corpora. Changing the native-reference corpus has only a limited effect on the correlations, whereas the overall correlation level differs between the two L2 corpora. For both tasks, LBS-4h native-reference corpus turns out to be the best choice. On S&I, distances calculated with cosine distance reach $\rho = -0.56$, while on UME-ERJ the strongest result ($\rho = -0.28$) is obtained with diagonal Mahalanobis distance. The weaker correlation on UME-ERJ is expected, since each recording contains only a short read sentence, leading to fewer observed phone-classes and possibly noisier utterance-level centroids. This consistency indicates that the effect is not tied to a single native-reference corpus.

Table 5 compares the proposed SVD-based L2 coordinates with the full feature space, using the original centroid space M_{nat}. By the Johnson-Lindenstrauss projection lemma [33], random projections should approximately preserve distances. The negative random-projection results serve as a sanity check that proficiency-related information is present in the proposed geometry. However, SVD is generally stronger than random projection and is close to the full space: it is best for Euclidean and Mahalanobis distances on S&I, and within 0.01 of the full space for Euclidean and Mahalanobis distances

Table 5: Spearman's ρ between distances and S&I Train proficiency scores or UME-ERJ learner pronunciation scores. We report Euclidean (Euc), Mahalanobis (Maha), and cosine distance metrics under three native-reference coordinate constructions: full (no projection), random (Rnd.) and SVD projections, both of rank 32. All experiments use triphone classes with full-unit averaging. More negative values indicate better agreement with proficiency. The best correlation in each row is highlighted.

L2 Task	Distance Measure	Spearman ρ		
		Full	Rnd.	SVD
S&I Train	Euc	-0.50	-0.46	$\mathbf{-0.52}$
	Maha	-0.44	-0.46	$\mathbf{-0.51}$
	Cosine	$\mathbf{-0.58}$	-0.53	-0.56
UME-ERJ	Euc	$\mathbf{-0.28}$	-0.26	-0.27
	Maha	$\mathbf{-0.29}$	-0.26	-0.28
	Cosine	$\mathbf{-0.19}$	-0.11	-0.13

Table 6: Final evaluation on S&I Dev and UME-ERJ learner using LBS-4h and UME-ERJ native-reference (Native-Ref) corpora. We report the range of matched phone-classes $|\mathcal{P}^\star|$, Pearson's r, raw and partial Spearman's ρ, for $|\mathcal{P}^\star|$ and silence ratio as control variables, and the 95% bootstrap confidence interval (Conf. Int.) for raw ρ. More negative values indicate stronger agreement with proficiency.

L2 Task	Native-Ref Corpus	$\lvert\mathcal{P}^\star\rvert$		Correlation				
					ρ			
		min	max	r	Raw	Partial	Conf. Int.	
							min	max
S&I Dev	LBS-4h	9	394	-0.53	-0.53	-0.19	-0.56	-0.49
UME-ERJ	UME-ERJ	2	54	-0.37	-0.34	-0.27	-0.37	-0.31

on UME-ERJ. Therefore, the SVD subspace provides a compact native-reference coordinate system that preserves the relevant acoustic and phonetic variation while reducing computation from 2520 to 8 seconds, corresponding to an approximately 315-times speedup.

As additional controls, replacing the self-supervised features with MFCCs substantially weakened the correlations, whereas constructing the native reference from a single TIMIT dialect group had only a minor effect.

3.5. Final Results and Discussion

In Table 6, we report both Pearson r and Spearman ρ correlation coefficients for a final evaluation using the best configuration described in Section 3.1. Given the high correlations of matched phone-classcoverage and silence ratio with the proficiency and pronunciation scores, shown in Tables 1 and 2, we additionally report Spearman correlations between distance and the scores while controlling for these variables. This is known as *partial correlation*, which measures the association between two variables after accounting for the effect of one or more control variables [34]. On S&I Dev, the native-reference distance shows a clear negative association with proficiency scores ($r = -0.53$, $\rho = -0.53$). After controlling for matched phone-classes and silence ratio, the correlation is reduced to $\rho = -0.19$ but remains negative, suggesting that the distance implicitly encodes these factors while also capturing additional variation. On UME-ERJ, despite much shorter read utterances and fewer matched phone-classes, the distance still correlates with pronunciation rating scores ($r = -0.37$, $\rho = -0.34$, partial $\rho = -0.27$). For both datasets, all reported correlations are statistically significant, with their 95% bootstrap confidence intervals lying entirely below zero. As a reference, listener-rated comprehensibility shows only moderate correlations with overall oral proficiency ($\rho = -0.260$ and -0.345) [35]. Therefore, $|\rho| \approx 0.3$ is meaningful for human-rating associations, and $|\rho| \approx 0.5$ is strong for our proposed method. In contrast, supervised S&I systems reach about $\rho = 0.83$ [23], but are trained on proficiency labels and can use information beyond pronunciation. We used the UME-ERJ task as a complementary evaluation condition to examine whether the proposed native-reference phone-class geometry transfers to a read-aloud setting. However, UME-ERJ short sentences provide fewer phone-class observations per utterance, which may make the utterance-level centroids less stable and the resulting distances less representative. Moreover, each learner utterance is rated by up to five raters, with an average pairwise human-human Spearman correlation of ≈ 0.50. Nevertheless, these results support that our proposed method captures information related to proficiency and pronunciation in spontaneous and read-aloud settings.

3.6. Limitations

In S&I, we evaluate spontaneous speech using holistic proficiency ratings. However, pronunciation, fluency, and phonetic coverage are interrelated, making it difficult to attribute the observed association specifically to pronunciation. We partly address this limitation by also evaluating on UME-ERJ, where the human ratings specifically target pronunciation quality, although the speech is read aloud. The contribution of other speech-related factors is also reflected in the partial-correlation analysis in Table 6, where controlling for matched phone-class coverage and silence ratio substantially reduces, but does not eliminate, the negative association. There is also no theoretical guarantee that the phone-class coordinates contain only phonetic information, as the underlying representations may encode other factors, such as speaker-dependent characteristics (e.g., accent) or recording conditions. Finally, although the proposed distance does not use acoustic-model likelihoods or posterior probabilities, it still requires phone-level segmentation obtained through forced alignment and therefore ASR transcripts.

4. CONCLUSIONS AND FUTURE WORK

In this work, we proposed a native-reference phone-class geometry for measuring L2 pronunciation deviation using self-supervised speech representations. By projecting the utterance-level L2 phone-class coordinates into a native-reference subspace, we computed distances between matched phone-classes and their corresponding native coordinates. We then showed that the proposed distances have a consistent negative correlation with speakers' proficiency in S&I and with learners' pronunciation in the UME-ERJ task, yielding Spearman's ρ of -0.53 and -0.34, respectively. The evaluations covered three native-reference corpora, spontaneous and read-aloud L2 tasks, and comparisons between different native-reference projection spaces. Using partial correlation analysis and confidence intervals, we confirmed that the negative association between the proposed distances and the L2 proficiency level is statistically supported. Unlike other methods, our proposed approach does not require pronunciation labels, read-aloud prompts, or matched native and L2 recordings of the same text. Even though our method relies on phone segmentations, we showed that under the best configuration, the correlation remains reasonably robust to random perturbation of the phone boundaries up to 60 ms. Future work will investigate how the proposed distances can complement the current state-of-the-art assessment systems as an additional pronunciation-related feature.

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